

Ashley Sample Management Agent - Project Summary Report

Document Info

- **Project Name:** Ashley Sample Management Agent (Pre-PLM AI Agent)
- **Report Date:** 2026-06-17

1. Project Overview

1.1 Background

In the R&D sampling process, product information management faces issues such as scattered information, entry limitations, inconvenient queries, and insufficient automation. This project aims to build a **low-barrier, real-time efficient** sampling information management assistant through Feishu Bot + AI Agent technology.

1.2 Goals & Achievement Status

Goal	Status	Description
Product Information Entry	 Complete	Intent recognition, auto field extraction
Multimodal Image Processing	 Complete	5 image types auto-classification
Auto PPT Report Generation	 Complete	4 versions (go_apply/trip_note/factory/all)
Product Query & Search	 Complete	Single product details + range search
Image Management	 Complete	View, delete, cloud storage
Fuzzy Entry	 Complete	Auto-generate placeholder numbers, can add later

2. Project Progress

2.1 Milestone Review

Phase	Time	Main Achievements
V2.0 Requirements Analysis	2026/5/19	Input/output modes, xwork-PLM interaction
Requirements Review	2026/5/20	Basic solution confirmed, HTML display postponed
xwork Integration	2026/5/20 - 5/21	Add business tables in xwork-PLM, interface integration
Version Development	2026/5/22 - 5/27	Basic functionality development complete
First Review	2026/5/28	PPT needs new versions, series/SKU distinction
Review Issue Fixes	2026/5/28 - 5/29	Multi-version support, clarify series/SKU concepts, refactor code
Architecture Optimization	2026/6/1 - 6/3	PPT generation optimization, token cost optimization, architecture simplification
Sampling Issue Fix 1	2026/6/5	Image extraction issues, multi-info queue mechanism
Sampling Issue Fix 2	2026/6/9	Add image type confirmation, optimize disconnection issues
WebSocket Fix	2026/6/10	Found probabilistic disconnection
Second Review	2026/6/11	Fuzzy entry, image view & delete
Review Issue Fixes	2026/6/11	Add no-series_no entry, image operations
WebSocket Fix 2	2026/6/12 - 6/15	Underlying fixes and intensive testing
Release & Documentation	2026/6/16 - 6/17	Output project documentation

2.2 Current Status

✔ Completed Features:

- Product entry/modification (text + image + file)
- 9 intent types recognition (entry, modify, query, search, report, image view, etc.)
- 4 PPT version generation
- 5 image type management
- Context memory and pronoun resolution
- Optimistic lock concurrency protection
- Dual-model adaptive routing (80% token cost savings)

🔄 In Progress:

- Editable H5 page (planned)

3. Technical Highlights

3.1 Architecture Innovation: Rule + LLM Hybrid Decision



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Layer 3: LLM Agent
(Intent Recognition + Tools)
- Tool Calling
- 9 intent categories
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Advantages:

- High-frequency commands respond in seconds, independent of LLM
- Core functions still work when LLM interface fails
- Token cost reduced by ~80%

3.2 Enhanced Intent Recognition

User Input	Recognized Intent	Tool Called
"How are today's samples"	Generate today's report	generate_ppt(note_date="today")
"How is UE0890"	Query single product	query_product
"Change its size to 30x20"	Modify previous product	preview_save(is_correction=true)
"Longwei sofa's number is UE0890"	Add formal number	assign_series_no

3.3 Multimodal Smart Processing

- **Auto Image Classification:** Product / Note / Material / Rendering / Problem
- **Problem Image Extraction:** Auto-identify annotated issues in images
- **Information Normalization:** Auto-clean formats for dimensions, prices, etc.

4. Bottlenecks & Solutions

4.1 Technical Bottlenecks

Bottleneck	Impact	Solution
High Token Cost	Operational cost pressure	Mini/Main dual-model strategy, optimize design, remove unnecessary skill/MCP design

Cost	Pressure	Design/Remove unnecessary skinner design
WebSocket Disconnection	Poor UX, probabilistic disconnection	Investigated AI code, found underlying logic issue

4.2 Process Bottlenecks

Bottleneck	Impact	Solution
Delayed Communication	Slowed progress, affected features	Reduce communication cost, constant alignment, detailed documentation for all issues
Workflow Integration	Slowed progress	Thorough research and understanding upfront

4.3 To-Do Items

Item	Priority	Status
Editable H5 Page	P2	Planned
Image Selection	P3	Planned

5. Lessons Learned

5.1 Success Factors

- Rule + LLM Hybrid Architecture**
 - Use rules for high-frequency scenarios, LLM for complex ones
 - Ensures response speed and cost control
- Confirmation Card Mechanism**
 - All write operations preview before confirm
 - Avoids data errors from AI misunderstanding
- Incremental Feature Release**
 - Core features first (entry/query), then extensions (PPT/images)

- Fast iteration, timely feedback collection

5.2 Lessons & Reflections

1. **Communication Must Be Thorough and Timely, Document Everything**

- Align requirements with business side, form documentation
- Think through entire project requirements comprehensively

2. **AI-Generated Code Needs Review**

- AI-generated code may have issues. For example, this time a small WebSocket underlying issue caused changes in other areas, affecting project progress.
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